

ISutra Procurement Officer & User Workflow Guide

A Practitioner's Manual for Drafting Compliant, Audit-Proof Tender Specifications

Target Audience: Public Procurement Officers, PSU Tender Committees, Municipal Engineers, and Smart India Hackathon (SIH) Evaluators

Introduction & Problem Statement

Public procurement in India must adhere to **Rule 144(i) of the General Financial Rules (GFR, 2017)** and the **Bureau of Indian Standards Act, 2016**. These statutory regulations dictate that all technical requirements and tender schedules must cite active Indian Standards (IS) wherever available.

However, procurement officers face recurring challenges:

- Manual Inefficiencies:** Sifting through thousands of standards is time-consuming.
- Obsolescence:** Copying text from past tenders introduces superseded standard numbers or withdrawn editions.
- Audit Disallowances:** The Central Vigilance Commission (CVC) and Comptroller and Auditor General (CAG) penalize tenders that cite ambiguous, proprietary, or unverified clauses.
- Linguistic Fragmentation:** Tender notices received from state departments in Hindi or Telugu cannot easily be mapped to English BIS catalogs.

ISutra acts as an intelligent decision-support copilot that transforms unstructured procurement requirements into verified, audit-grade Indian Standards recommendations.

The Complete 7-Step Officer Workflow



Step 1: Input Tender Requirements

Navigate to the **Analyze** page ([AnalyzePage.tsx](#)).

ISutra supports two input modalities:

Option A: Free-Text Specification

Paste technical specification clauses, tender items, or descriptions directly into the text editor. You can draft in:

- **English:** Standard engineering and procurement terms.
- **Hindi (हिंदी):** e.g., "120W आउटडोर एलईडी स्ट्रीट लाइट ल्यूमिनेयर, IP66 जलरोधी, 10kV सर्ज प्रोटेक्शन".
- **Telugu (తెలుగు):** e.g., "120W అవుట్‌డోర్ LED (స్ట్రీట్ లైట్ లూమినైర్, IP66 రక్షణ, 10kV సర్జ్ ప్రొటెక్షన్".

The system's Indic translation layer ([multilingualService.ts](#)) automatically normalizes regional terminology to canonical engineering classifications while preserving the original source text snippet for audit trails.

Option B: Document Ingestion

Drag and drop digital tender schedules:

- Supported formats: `.pdf` , `.docx` , `.doc` (up to 15 MB).
- The system verifies the internal file signature (magic bytes) to ensure file security and parses text streams with structure preservation.

NOTE

****Scanned Document Handling**:** If an uploaded PDF contains only flattened images/scans without a digital text layer, the system returns an informative notice ('422 Unprocessable Entity') explaining that the document is scanned and advising the user to upload a digital export or enable server-side OCR.

Step 2: Human-in-the-Loop Review Barrier

After ingestion, the system loads the **Requirement Review Interface** (`RequirementReviewPage.tsx`).

Why Human Review is Mandatory:

In government procurement, an autonomous AI must never be allowed to bind a statutory tender contract. The AI acts as an **accelerator of perception**, extracting 20+ dimensions:

- Product Name & Taxonomy (Category, Subcategory)
- Application Domain (Highway, Residential, Industrial, High-Mast)
- Operating Environment (Outdoor, Marine/Coastal, High-Ambient Temp)
- Technical Parameters (Wattage, Voltage, Efficacy, CCT, Conductor Size)
- Safety & Ingress Ratings (IP Rating, Surge Protection, Fire Retardance)
- Mandatory Testing Protocols (Type Tests, Routine Tests)

Officer Actions in the Review Interface:

1. **Verify Values:** Review extracted values against your original tender draft.
2. **Edit Parameters:** Click any field to update values (e.g., change wattage from 120W to 150W).
3. **Add Missing Parameters:** Use the "+ Add Parameter" button to supply criteria that the tender text omitted.
4. **Flag Uncertainty:** Mark any parameter with "**Needs Review**" if clarification is required from field engineers.
5. **Confirm Requirements:** Click "**Confirm & Find Matching Standards**".

IMPORTANT

The recommendation engine matches ****strictly and exclusively**** against the confirmed parameters. Unchecked AI outputs are never sent to the decision engine.

Step 3: Evaluating Recommendations & Explainability

On the **Recommendations Page** (`RecommendationsResultsPage.tsx`), standards are ranked by relevance.

Understanding the 6-Factor Deterministic Score:

Every score ($S \in [0.0, 1.0]$) is calculated by a weighted linear combination across six signals:

Factor	Weight	What the Engine Validates
Product / Category	30%	Does the standard govern this exact equipment family (e.g., Luminaire vs. Cable vs. Switchgear)?
Keywords, Title & Scope	25%	Do the title and scope statement in official BIS publications explicitly cite the requested terminology?
Application Domain Fit	15%	Is the standard designed for the intended installation (e.g., Roadway/Street lighting vs. Mine lighting)?
Environmental Context	10%	Does the standard contain clauses covering outdoor, dust, water, or thermal extremes?
Technical Parameters	10%	Does the standard specify electrical limits, mechanical ratings, or conductor cross-sections?
Safety & Testing	10%	Are type tests, ingress protection tests, and insulation resistance tests specified?

Specificity Tiers (Tiers 1 to 6):

ISutra prevents generic standards (such as general test methods or component codes) from outranking specialized equipment codes:

- **Tier 1 (Exact Product & Application Match):** Primary standard directly governing the equipment.
- **Tier 2 (Close Subcategory Match):** Closely related standard in the same subcategory.
- **Tier 3 (Broader Equipment Family):** General standard covering the broader equipment group.
- **Tier 4 to 6 (Allied / Component / General Test Method):** Normative reference or component-level code.

"Why This Standard?" Deep Dive:

Click the **"Why This Standard?"** button on any card to reveal the factor point breakdown:

- View the exact points earned (e.g., Product Category: `+0.300 / 0.300` , Application Domain: `+0.150 / 0.150`).
- Review the specific evidence phrases extracted from the standard's text.
- Inspect the unbroken **5-Stage Audit Trail**: `User Requirement` $\rightarrow$ `Extracted Parameter` $\rightarrow$ `Scoring Signal` $\rightarrow$ `Standard Clause` $\rightarrow$ `Official BIS URL` .

Step 4: Requirement Gap Analysis

A common cause of tender litigation is assuming a single Indian Standard covers every requirement in a procurement schedule.

Click "**Run Gap Analysis**" on a standard to enter the **Requirement Gap Analysis Page** ([RequirementGapAnalysisPage.tsx](#)).

The 4-State Reference Coverage Taxonomy:

Every confirmed parameter is audited against the standard's verified clauses:

1. **supported (Green)**: The parameter is explicitly governed and tested within this standard (e.g., Ingress Protection IP66 under IS 10322 Part 5 Sec 3).
2. **not_supported (Red)**: The parameter is contradicted or excluded by the standard (e.g., specifying 33kV for a standard restricted to 1.1kV).
3. **not_available (Amber)**: The standard does not govern this dimension (e.g., safety standard IS 10322 does not prescribe photometric efficacy in lm/W; that is governed by performance standard IS 16107 Part 2 Sec 1).
4. **needs_verification (Blue)**: The clause exists in the standard but requires officer verification against specific project schedules or manufacturer type-test certificates.

CAUTION

****Statutory Notice****: ISutra strictly labels this matrix as ****Reference Coverage**** and explicitly disclaims issuing a "Compliance Certificate". Only BIS-licensed laboratories can certify physical product compliance.

Step 5: Allied, Normative & Testing Standards

Complex equipment tenders require citing complementary standards to ensure comprehensive quality assurance.

Access the **Allied Standards Explorer** ([StandardDetailsPage.tsx](#)):

- **Normative References**: Compulsory parent standards referenced in Clause 2 of the primary standard.
- **Test Methods**: Standards governing physical, chemical, or electrical laboratory testing (e.g., IS 10810 series for cable testing).
- **Safety Standards**: Fundamental electrical safety or fire retardancy codes (e.g., IS/IEC 60529 for Degrees of Protection).
- **Installation Standards**: National Electrical Code (NEC) or CPWD guidelines for civil installation and earthing.

Each relationship displays an **Evidence Clause** and direct hyperlinks to official BIS gazette records, ensuring no relationship is fabricated.

Step 6: Lifecycle Timeline & Mandatory QCO Verification

Before issuing a tender, officers must ensure they are not citing withdrawn standards and that mandatory certification orders are satisfied.

Visual 5-Stage Lifecycle Stepper:

The system tracks the evolution of each standard: $\text{\$}\backslash\text{Base Standard}\backslash\longrightarrow\backslash\text{Revision (Rev 1, 2...)}\backslash\longrightarrow\backslash\text{Amendments}\backslash\longrightarrow\backslash\text{Reaffirmation}\backslash\longrightarrow\backslash\text{Official Gazette}\backslash\text{\$}$

- **Reaffirmation Tracking:** Standards are reviewed every 5 years by BIS sectional committees. ISutra shows the latest reaffirmation year (e.g., IS 456 reaffirmed in 2025).
- **Gazette Amendment Tracking:** Displays all published amendments with effective dates.

Regulatory Intelligence (Quality Control Orders):

Under the BIS Act 2016, Ministries publish mandatory **Quality Control Orders (QCOs)** in the Gazette of India:

- **VERIFIED APPLICABILITY (Mandatory):** Products under this standard (e.g., Steel TMT bars under IS 1786, or Footwear under IS 15844) cannot be legally manufactured, imported, or procured without a valid BIS Standard Mark (ISI mark).
- **VERIFICATION REQUIRED :** The standard is subject to an upcoming or phased QCO enforcement window.
- **NO VERIFIED REGULATORY RECORD :** Standard operates under voluntary certification unless mandated by specific departmental tender rules.

Step 7: Exporting Audit-Grade Intelligence Reports

Once the review and standard selection are finalized, click "**Generate Procurement Report**".

Printable Evaluation Report Features:

- **Clean Government Header:** Formatted for institutional filing and committee reviews.
- **Full Traceability Matrix:** Proves to vigilance and audit authorities that standards were selected objectively through mathematical criteria.
- **Identified Specification Gaps:** Lists allied standards that must be added to the tender to cover gaps in the primary code.
- **Direct Gazette Links:** Contains verified <https://services.bis.gov.in> and <https://egazette.gov.in> hyperlinks for instant officer verification.
- **Exporting Options:** Click "**Print / Save PDF**" to save an audit dossier directly into your official e-Office or tender management folder.

